

# Ahmad AlAmmouri

STAFF RESEARCH ENGINEER · WIRELESS COMMUNICATIONS

Plano, TX, USA

## Summary

☎ (+1) 737-333-5428 | ✉ alammouri.ahmad@gmail.com | 🌐 www.alammouri.com | 📱 ahmad-alamouri

Staff Research Engineer with a PhD in Wireless Communications from UT Austin and over a decade of experience designing and optimizing next-generation wireless systems, including mmWave and FR3. Led system-level architecture, MIMO design, and hybrid beamforming development for multiple PoCs, bridging theory and real-world deployment constraints. Co-inventor on 20+ patents and author of 25+ publications. Experienced in driving end-to-end system design, simulation, and cross-functional execution.

## Research Experience

### Samsung Research America (SRA)

Plano, Texas

#### STAFF RESEARCH ENGINEER

Mar. 2024 - Now

- Led system-level design of mmWave FWA solutions, developing air-interface algorithms and beamforming strategies that increased supported CPEs per access unit by up to 117% in system-level evaluations..
- Designed and implemented Joint Phase-Time Array (JPTA) PoC for mmWave/THz systems, enabling uplink coverage extension through hybrid beamforming and time-domain processing.
- Owned system-level design for FR3 PoC, defining MIMO architecture and feature set for hybrid beamforming under hardware, deployment, and link budget constraints.
- Drove system-level link budget analysis and feature planning for FR3 evaluation, enabling performance assessment under realistic propagation and hardware assumptions.
- Led cross-team coordination across three Samsung research hubs, aligning system architecture, feature roadmap, and PoC execution for JPTA, FR3, and FWA initiatives.
- Designed energy-aware multi-device connectivity framework, enabling cooperative communication strategies across devices to improve system efficiency under power constraints.
- Contributed to AI-RAN system design by proposing a delay-aware architecture with hierarchical agents and orchestrators, enabling scalable coordination and latency-aware decision-making in dynamic wireless networks.
- Led investigation of UL-MIMO design for FR3 systems, evaluating codebook-based vs non-codebook-based and coherent vs non-coherent transmission schemes, and proposed multi-TRX FDM approaches to improve system flexibility and performance under practical deployment constraints.

#### SENIOR RESEARCH ENGINEER

Apr. 2021 - Feb. 2024

- Designed and implemented Joint Phase-Time Array (JPTA) PoC for mmWave/THz systems, enabling frequency-selective analog beams and 8x or more UL scheduling opportunities for multi-user scenarios..
- Developed MU-MIMO strategies for upper-mid band systems, including scheduling, hybrid precoding, and 3GPP-compliant design tradeoffs.
- Designed beam management and codebook strategies for Samsung 5G mmWave base stations under practical deployment constraints.
- Developed and maintained system-level, link-level, and ray-tracing simulators to evaluate beamforming and multi-user performance across realistic channel conditions.

### National Institute for Research in Digital Science and Technology (INRIA)

Paris, France

#### CHATEAUBRIAND RESEARCHER

Jan. 2020 - Jul. 2020

- Data Traffic Dynamics in Cellular Networks.

#### VISITING RESEARCHER

Apr. 2019 - Jul. 2019

- Data Traffic Dynamics in Cellular Networks.

### Samsung Research America (SRA)

Richardson, Texas

#### RESEARCH INTERN

May 2018 - Aug. 2018

- Beam Codebook Design for 5G Mobile Devices With the Impact of User's Handgrip.

#### RESEARCH INTERN

May 2017 - Aug. 2017

- Integrated Access and Backhaul for 5G mmWaves Cellular Networks.

## Education

### The University of Texas at Austin

#### PHD IN ELECTRICAL ENGINEERING

- Advisors: Prof. Jeffrey Andrews and Prof. François Baccelli
- Thesis: Analysis of Cellular Networks: Densification and Data Traffic Dynamics
- GPA: 4.00/4.00
- A technical reviewer for 50+ peer-reviewed journal papers and the recipient of three Exemplary Reviewer awards for that.
- A technical program committee (TPC) member for ICC 2018-2024.

Austin, Texas

Aug. 2016 - Jan. 2021

### King Abdullah University of Science and Technology (KAUST)

#### MS IN ELECTRICAL ENGINEERING

- Advisor: Prof. Mohamed-Slim Alouini
- Thesis: Full-Duplex Communications in Large-Scale Cellular Networks
- GPA: 4.00/4.00

Thuwal, Saudi Arabia

Aug. 2014 - May 2016

### The University of Jordan

#### BS IN ELECTRICAL ENGINEERING

- Graduation Project: Distributed Opportunistic Spectrum Sharing in Cognitive Radio Networks
- GPA: 3.73/4.00, Valedictorian, Excellent with Honor

Amman, Jordan

Aug. 2009 - Jan. 2014

## Technical Skills

**Wireless Standards:** 3GPP NR (PHY-layer), ORAN.

**Programming:** MATLAB, Python, C++, Git.

**Tools:** SLS, LLS, PyTorch, Remcom Wireless InSite, HFSS.

**Theoretical Foundations:** Signal Processing, Beamforming, MU-MIMO, Hybrid Precoding, Channel Estimation, Stochastic Geometry, Queuing Theory.

## Honors & Awards

- |            |                                                                                                                                      |
|------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 2022-2025  | <b>SRA Inventor Award</b> , High quality patents for X-MIMO and JPTA System Development.                                             |
| 2023       | <b>SRA Individual Award</b> , Projects: JPTA and X-MIMO System Development.                                                          |
| 2021, 2023 | <b>SRA Team Award</b> , Project: JPTA System and PoC Development.                                                                    |
| 2020       | <b>Student Leadership Award</b> , awarded by the WNCG at UT Austin.                                                                  |
| 2019       | <b>Chateaubriand Fellow</b> , awarded by the Embassy of France in the United States.                                                 |
| 2019       | <b>Professional Development Award</b> , awarded by the ECE department at UT Austin.                                                  |
| 2017, 2018 | <b>Exemplary Reviewer</b> , IEEE Transactions on Wireless Communications.                                                            |
| 2018       | <b>SRA President's Award</b> , Beam Codebooks for mmWaves Mobile Devices.                                                            |
| 2017       | <b>Exemplary Reviewer</b> , IEEE Transactions on Communications.                                                                     |
| 2016       | <b>Best Poster Award</b> , KAUST-NSF Conf. on Electronic Systems for a Sustainable Future.                                           |
| 2014-2016  | <b>King Abdullah University of Science and Technology Fellowship</b> , awarded to only one student from Jordan in the class of 2016. |
| 2014       | <b>University of Jordan Award for Scientific Excellence</b> , top of Electrical Engineering class.                                   |

## Selected Publications and Patents

**1 [B] Book. 14 [J] Journal paper. 12 [C] Conference paper. 21 [P] Patents.** <Full list available here>

**[B1]** J. G. Andrews, A. K. Gupta, **A. AlAmmouri**, and H. S. Dhillon, "An Introduction to Cellular Network Analysis using Stochastic Geometry", Springer, Jun. 2023.

**[J4]** **A. AlAmmouri**, J. MO, V. V. Ratnam, B. L. NG, R. W. Heath, J. Lee, J. Zhang, "Extending Uplink Coverage of mmWave and Terahertz Systems Through Joint Phase-Time Arrays", IEEE Access, Aug. 2022.

**[J3]** **A. AlAmmouri**, J. Mo, , B. L. Ng, J. C. Zhang, and J. G. Andrews, "Hand Grip Impact on 5G mmWave Mobile Devices", IEEE Access, May 2019.

**[J2]** **A. AlAmmouri**, J. G. Andrews, and F. Baccelli, "A Unified Asymptotic Analysis of Area Spectral Efficiency in Ultradense Cellular Networks", IEEE Transactions on Information Theory, Jun. 2018.

**[J1]** **A. AlAmmouri**, H. ElSawy, O. Amin, and M. -S. Alouini, "In-Band  $\alpha$ -Duplex Scheme for Cellular Networks: A Stochastic Geometry Approach", IEEE Transactions on Wireless Communications, Jul. 2016.